# Title: Titanic Survival Prediction using Machine Learning

**1.Introduction:** This project focuses on predicting whether a passenger survived the Titanic disaster using machine learning techniques. It involves data preprocessing, visualization, and model building.

**2.Objective:** The objective of this project is to understand the complete data science pipeline including data cleaning, visualization, feature engineering, and model evaluation.

**3.Tools & Technologies:**

1.Python 2. Pandas

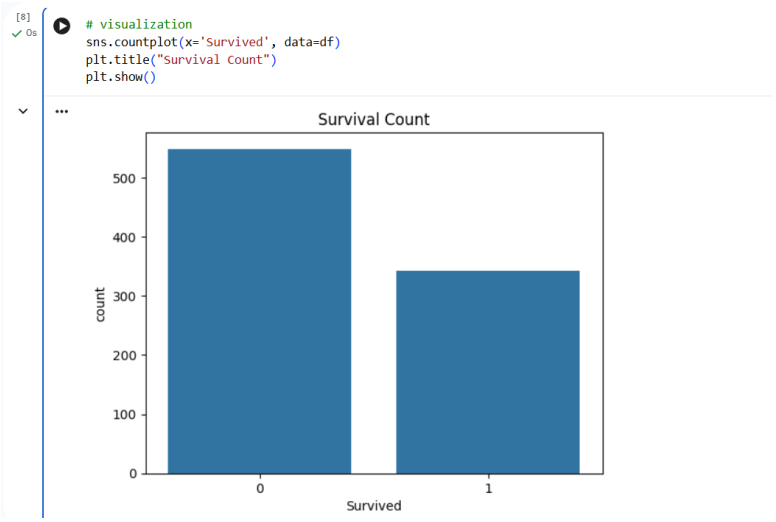
3.NumPy 4. Matplotlib

5.Seaborn 6. Scikit-learn

**4. Methodology:**

1. Data Collection – Titanic dataset loaded
2. Data Cleaning – Missing values handled
3. Data Visualization – Graphs created
4. Feature Engineering – Family Size added
5. Model Building – Logistic Regression used
6. Evaluation – Accuracy and confusion matrix calculated

**5. Data Visualization: i) Survival Count**

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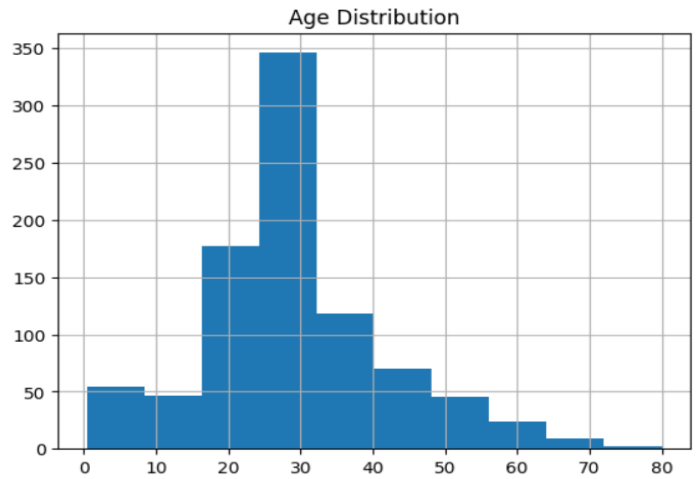
## ii) Age Distribution

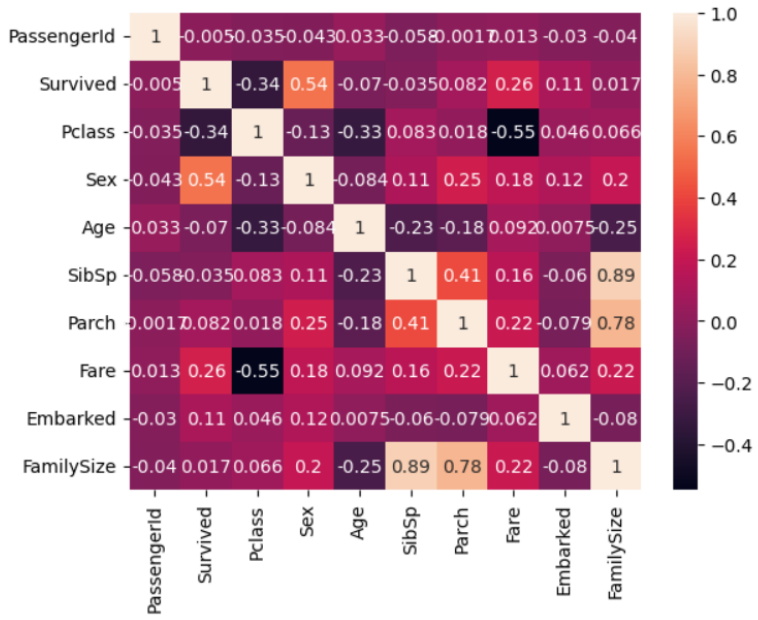
```
[9]
df['Age'].hist()
plt.title("Age Distribution")
plt.show()
```




## iii) Correlation Heatmap

```
[14]
sns.heatmap(df.select_dtypes(include=['number']).corr(), annot=True)
```

```
<Axes: >
```

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## 6. Model Evaluation:

```
[15]
X = df[['Age', 'Sex', 'Pclass', 'Fare', 'FamilySize']]
y = df['Survived']

[16]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

[17]
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

LogisticRegression
LogisticRegression(max_iter=1000)

[18]
y_pred = model.predict(X_test)

[21]
# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))

Accuracy: 0.8100558659217877
Confusion Matrix:
 [[104  14]
 [ 20  41]]
```

The model achieved an accuracy of 81%. The confusion matrix shows that most predictions are correct, indicating good model performance.

## 7. Results:

The model successfully predicts survival with good accuracy. Features like gender and passenger class have a strong impact on survival.

## 8. Conclusion:

This project demonstrates how machine learning can be applied to real-world data. It helps in understanding the importance of data preprocessing and model evaluation.

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